
THE EH ANTENNA – THE ULTIMATE PROOF

By Claudio Re, IIRFQ

Preface

Starting around from 1980 some claims of the possibility of generating a new category of antennas was proposed by Hatley, Kabbary, Zimmerman, Hart and others. Strangely none of the proponents were able to put together any mathematical theory or demonstration that survives a minimal check. Most of the claims told that a Poynting Vector Synthesis (PVS) would make it possible to build very short and efficient antennas. Some typically fuzzy experiments were given to prove these assumptions. As far as I know, none of the claims have survived any scientific investigations.

Among many others, I personally tested some of these antennas (EH, DLA, X3), finding that there was nothing about them that challenged the classical Maxwell theories. Something strange was noticed at first sight from non-clinical and critical tests. Such was typically due to common mode currents on the feeder lines. All these effects disappeared with true no-feeder tests or good precautions to minimize the effects of the common mode currents until they were negligible in the measurements. In the References at the end of the article, in contrast, usually one would find some good tests made by hams claiming use of scientific methods. Zimmerman and others were honest enough to admit the typical error of the common mode currents.

After having used the experimental way to check PVS. (On the EH there is a series of two articles, the first is available on the *antenneX* public library. See references at the end and **Table 1** with the summary of the results of the second article measurements.) One day I had the idea to see what can happen if I try to change the test approach. This was summarized in this question:

“What could theoretically happen combining the E and H fields produced from two different antennas?”

This question generated a fourth multi-part series of *antenneX* articles: “A new approach to Poynting Vector Synthesis,” quite fully based on math. The most elegant and general demonstration that PVS doesn’t exist was proposed from Alan Boswell and is included in the fourth part. This series of effort gave me a lot of satisfaction and pushed me to improve my knowledge of advanced math with the great help of the co-author Federico Vavassori, but the inconvenience is that too much deep math is a bit dry for some of the readers. Thinking of the way to make the “ultimate proof” comprehensible and interesting to the widest audience of *antenneX* I decided to perform a new approach to this issue in three steps:

- 1) Model with NEC an EH LT antenna and change the supposed radiator element with its equivalent circuit.
- 2) Do the same with the feeder cable.
- 3) Check if this system is in the range of results claimed from the EH supporters.

If the check is okay, it obviously means that the EH antenna is simply another Hertzian antenna.

Description	Fig.	Gain dBd	B RL MHz	Q RL	B1 FDT MHz	Q1 FDT
EH LT	19	-17.0	1.784	28.0	2.00	25.0
EH LT GP 40x100 cm	20	-15.8	0.166	301.2	0.16	312.5
Cone LC GP 20x50 cm	22	-19.3	0.250	200.0	0.25	200.0
Cone LC GP 40x100 cm	23	-21.2	0.184	271.7	0.16	312.5
Aluminium Loop 23cm GP	24	-9.2	0.416	120.2	0.50	100.0
EH Lattice	21	-16.7	1.916	26.0	1.50	33.3

Table1. Results of the gain measurements of various configurations of EH antennas and a small Loop compared with a Ground Plane at 50 MHz

First and second step

Figure 1 shows the circuit of the EH LT Bicone antenna that I have analyzed in the two part series of articles for *antenneX*.

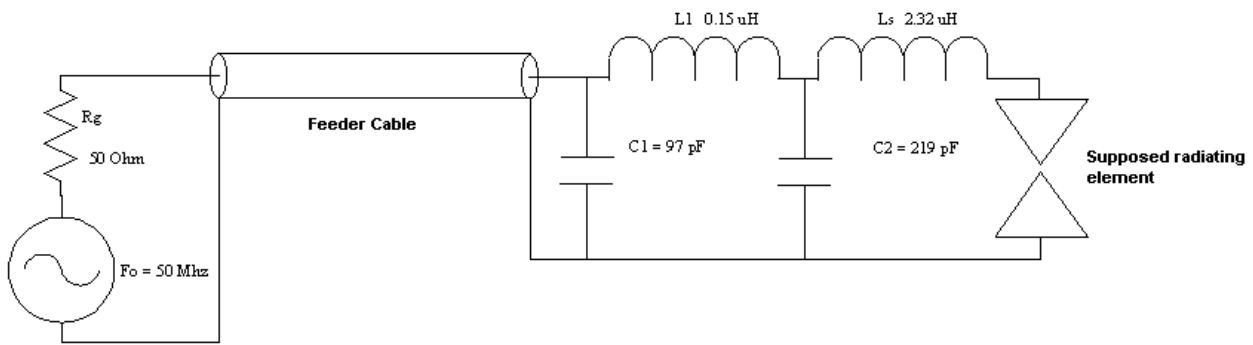


Fig 1. Circuit of the EH LT Bicone Antenna at 50 MHz.

We can now put, instead of the supposed radiating element, its equivalent circuit following the classical theory: a capacitor of 4.5 pF with negligible losses. We also add for the L_s a reasonable loss as a resistance of around $2\ \Omega$ in series (R_s). As a first approximation, let's skip the PI network that in the classical theory is intended as a matching network from low resistance ($10\ \Omega$ on the right side) to higher resistance ($50\ \Omega$ on the left side). The cable equivalent is an ideal cable with internal currents I_1 and I_2 and an outside shell where the unbalancing current I_3 can flow if we have common mode excitation.

We get the new circuit of **Figure 2**.

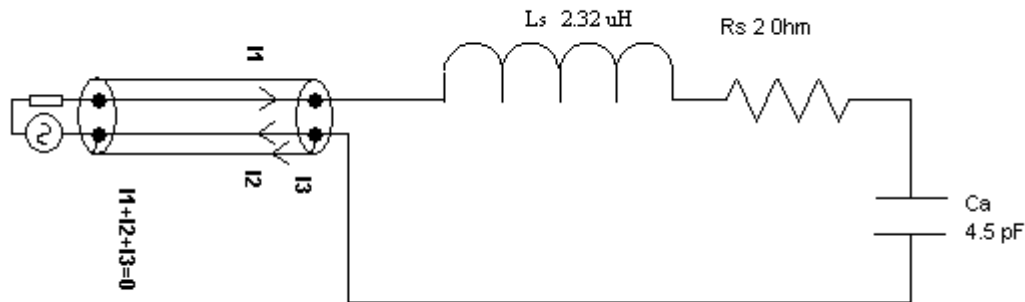


Fig. 2. Circuit of the EH LT Bicone Antenna at 50 MHz. Cable with the differential and common mode currents shown

The internal line of the coax is balanced. Considering negligible the losses in this line, there is no reason why we can't shift the generator all on the left of the cable. This is shown in **Figure 3**. The cable is now a tubular conductor.

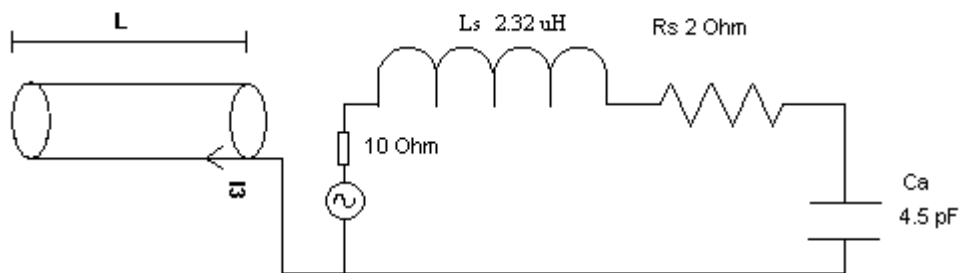


Fig. 3: Circuit of the EH LT Bicone Antenna at 50 MHz. The generator has been passed through the inner equivalent part of the cable

NEC Modelling

The circuit of **Figure 3** was modelled and analyzed around resonance with Multinec for various lengths of the cable. To properly insert the concentrated components, a square loop of 1cm was built, with the NEC generator and the L_s , R_s and C_a placed on the wires like shown in **Figure 4**. The shield of the cable was modelled with a wire.

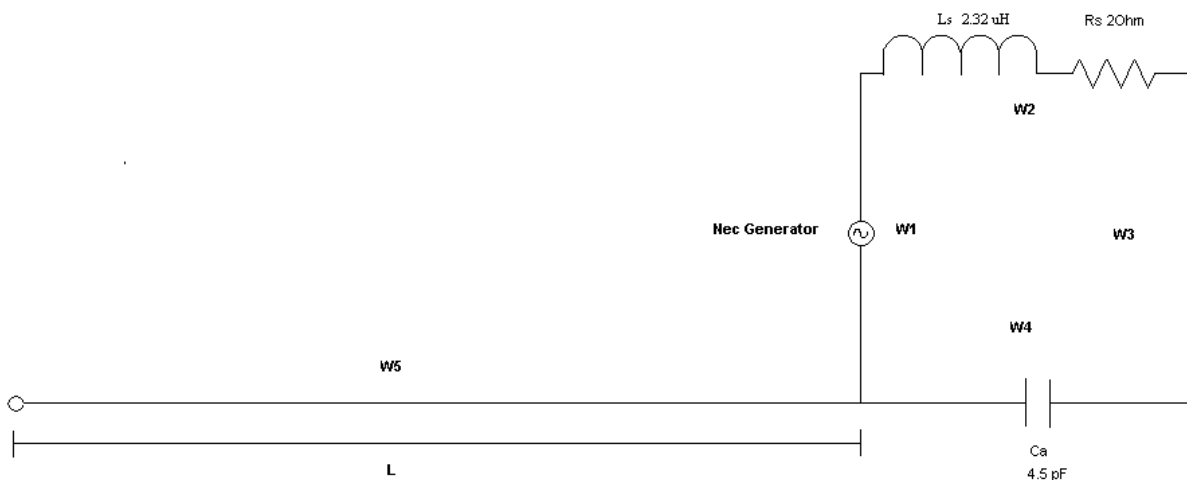


Fig. 4. Modelling the circuit of Figure 3 with Multinec

The antenna was tuned to resonance at 50 MHz with $C_a = 3.9$ pF. A length of 2.67 m (around $WL/2$), gave the 10Ω s needed for a perfect match. (Remember that we took out from the circuits the PI matching unit from 10 to 50 Ω s, so our Z_o is 10 Ω s.)

The results are shown in **Figure 5**.

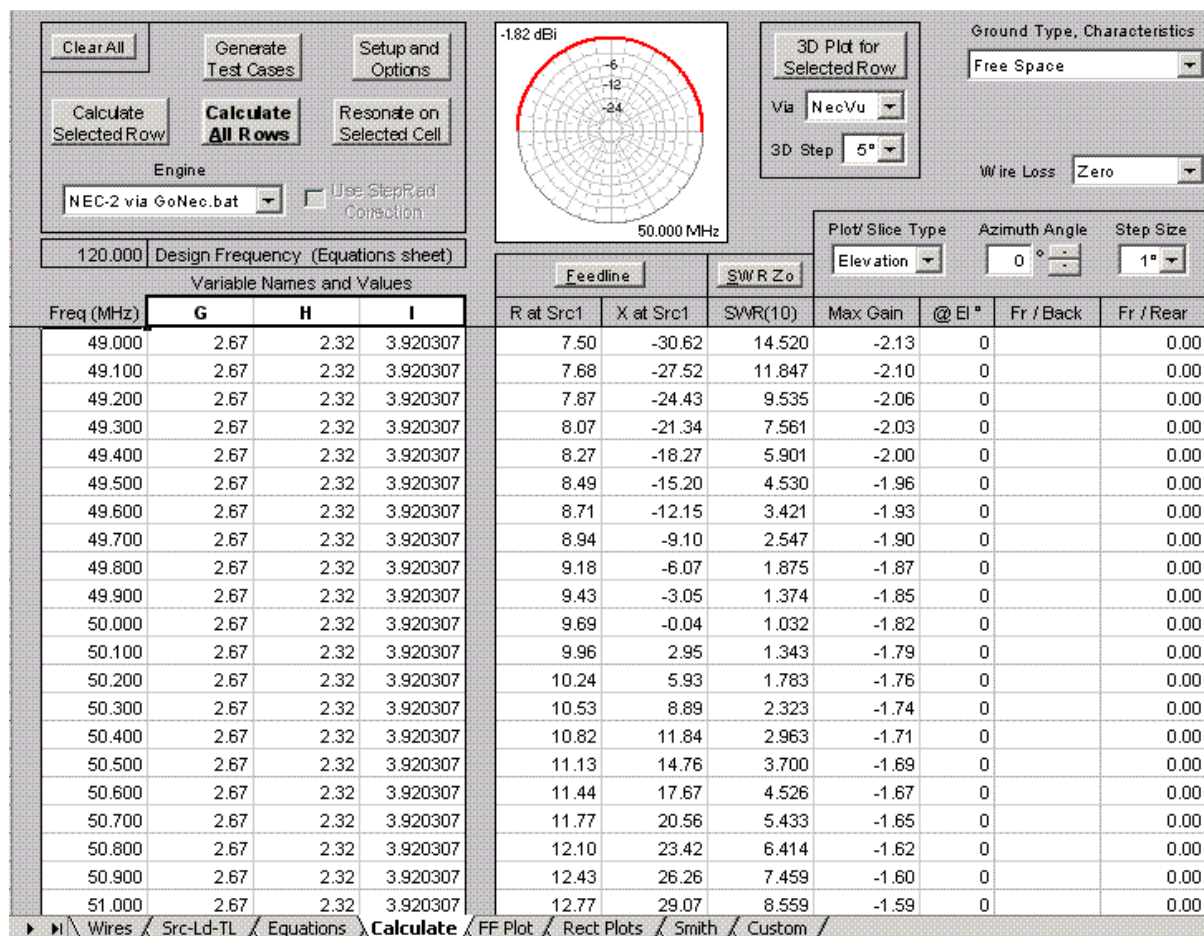


Fig. 5: Results of the Multinec simulation.

The gain is -1.8 dBi. The VSWR is plotted in **Figure 6**.

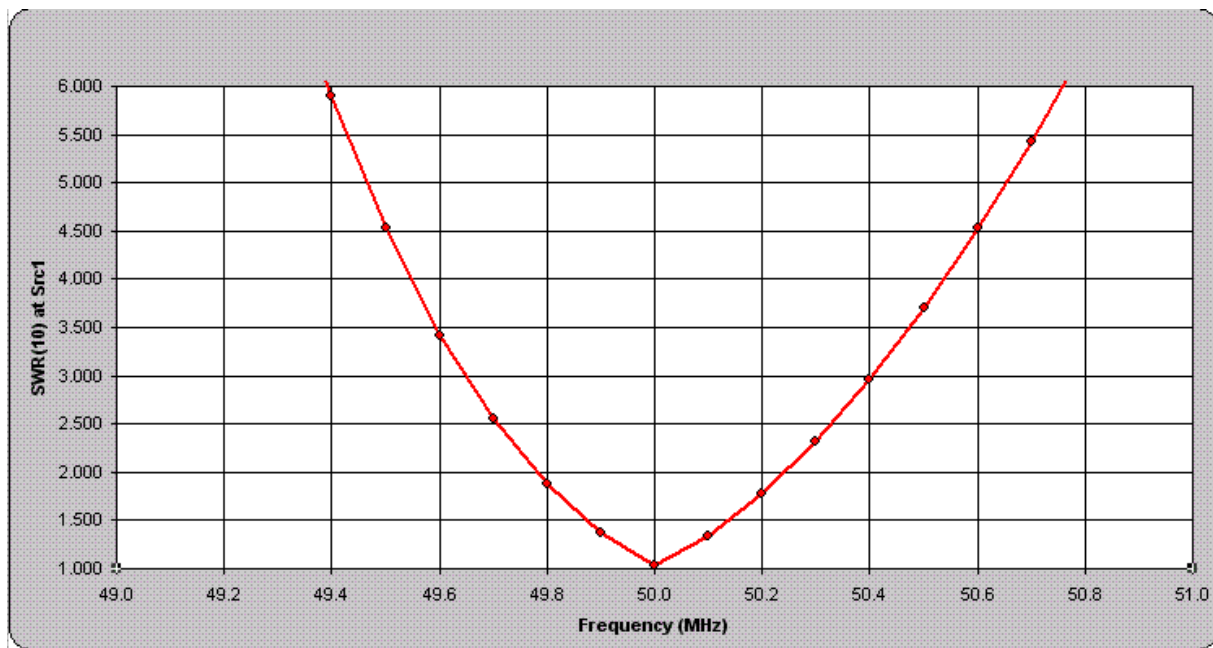


Fig. 6. VSWR plot of the Multinec simulation.

The Q of this antenna is around 88, computed with the points with VSWR = 2.6 with the formula:
 $Q = F_0/B$.

The coaxial in this ideal case is fully floating in free space. In practical cases, the environment where the cable lies will dump much more the antenna.

Conclusions

What have we learned?

Every short antenna is mainly a little capacitor. To resonate it, we need to add a big inductor. This forms a resonant circuit that easily inducts currents as common mode current on the shield of the coaxial cables that will radiate. Every circuit in common, inductive or capacitive, is able to do this, like in **Figure 3** where the common path is only 1 cm! For this it is not necessary to claim or invent new theories. As we have demonstrated with the use of Multinec, the conventional theories are sufficient.

Two additional proofs for dummies

Proof n.1

Just redrawing a little **Figure 4**, I think that everyone could recognize something very similar to a $WL/2$ vertical antenna excited on one side. This is shown in **Figure 7**. The L_s and C_a form a L circuit that match the high impedance of the end-fed $WL/2$ radiator.

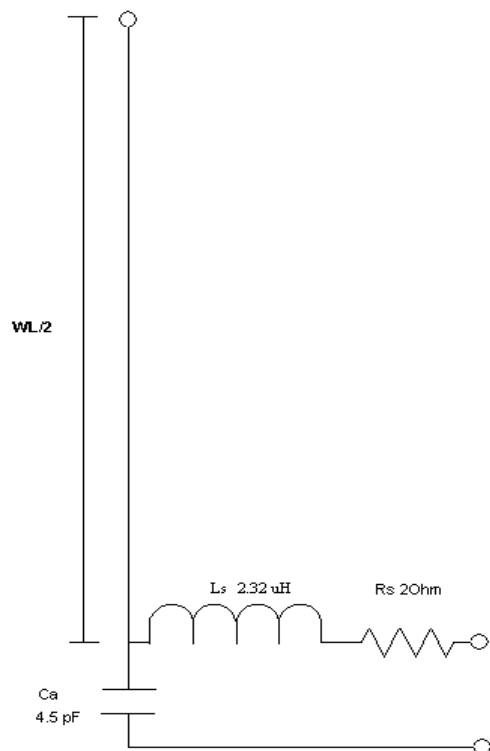


Fig. 7. The easiest way to understand how the antenna works.

Proof n.2

If we suppress the radiating part of the antenna (the cable) there is no way to radiate and to match the antenna (if not adding losses). This is demonstrated in the following **Figures (8 ,9, 10, 11, 12 and 13)** that are self-explanatory. No cable, no match. On the other hand, the EH is able to tune any component added to the floating circuit, like for example, one of my arms. A system to feed and tune everything is connected to this circuit. This is what the EH really is like many of the other short loaded antennas.



Fig. 8. Set-up of the EH LT Bicone at 50 MHz



Fig. 9. Set-up of the measurements with a “random” feed cable, like the EH is normally connected



Fig. 10. Measurement with the random feed cable



Fig. 11. Measurement made removing the cable (no feeder measurement): no match at all. $R_s = 5 \Omega$ due to coil loss and little radiation from the body of the MFJ 259. Prospective problems of the picture shows the cable of the first setup near to the top of the EH. Really it is more than 50 cm far from the EH.

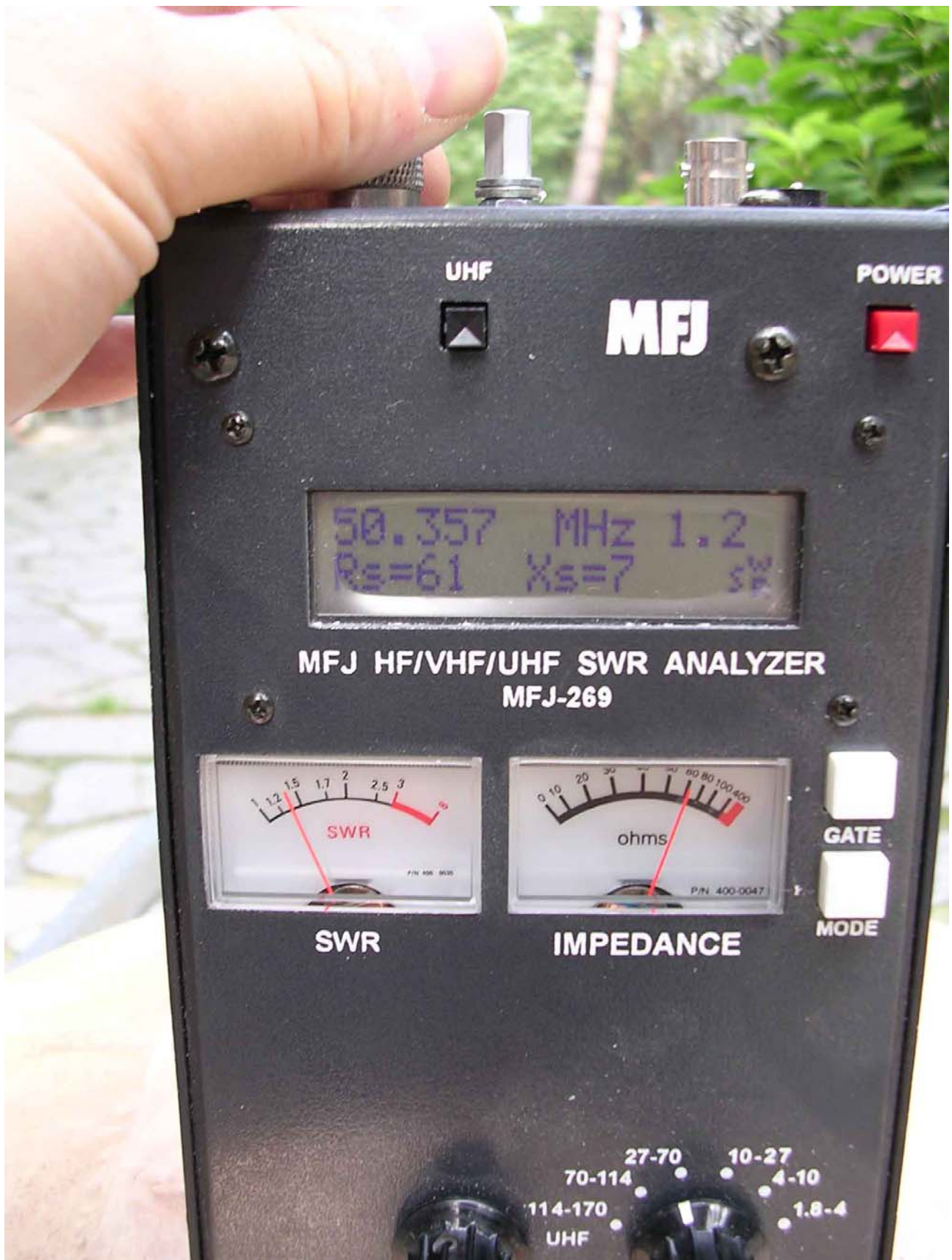


Fig. 12. Adding one of my hands on the body of the MFJ 269 and slightly changing the frequency it is possible to achieve again a good match. The system is tuning my lossy body.



Fig. 13. Expanded picture of Figure 12

References from: http://www.comsistel.com/nuova_pagina_2.htm

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- 2) EH ionosferic gain comparison at 14 MHz with a Ground Plane: EH1.GIF
- 3) EH gain summary of measurements at 50 MHz: EHgain.xls
- 4) http://home.earthlink.net/~calvinf15/technical/EH_antenna_test_report.pdf
- 5) http://home.earthlink.net/~calvinf15/technical/EH_antenna_test_report_2.pdf
- 6) http://www.eh-antenna.fromru.com/meas_0.html
- 7) <http://www.eh-antenna.fromru.com>
- 8) IEEE Article on tests on Short Medium Frequency Antennas and CFA - V.Trainotti.pdf

BRIEF BIOGRAPHY OF AUTHOR

Claudio Re, I1RFQ~ [Email](#)

Claudio Re is a graduate of Polytecnic of Turin in 1980 with specialization in Telecommunications and Hyperfrequencies. He is an owner of Sistel SRL and Sinfotel SRL, two small telecommunications companies in Italy. He is also a consultant and the Network Director of the World Family of Radio Maria, a Catholic Broadcasting Network that broadcasts now in 27 different countries of the globe.

Claudio has experimented with all kinds of equipment from 136 KHz to the optical frequencies (one-way communications at 22km). He was born in 1956, built his first crystal set receiver at age 6 and at 16, obtained his license with the callsign I1RFQ.



Claudio Re, I1RFQ

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